

Math 242, S13
Quiz 8

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find the general solution of the following homogeneous second-order differential equations:

(a)

$$y'' + 8y' + 25y = 0$$

Solution: characteristic equation:

$$r^2 + 8r + 25 = 0$$

$$\Rightarrow r = \frac{-8 \pm \sqrt{8^2 - 4 \cdot 1 \cdot 25}}{2} = \frac{-8 \pm 6i}{2}$$
$$= -4 \pm 3i$$

$$\Rightarrow y(x) = e^{-4x}(C_1 \cos 3x + C_2 \sin 3x)$$

(b)

$$9y^{(3)} + 12y'' + 4y' = 0$$

Solution: characteristic equation:

$$9r^3 + 12r^2 + 4r = 0$$

$$r(9r^2 + 12r + 4) = 0$$

$$r(3r + 2)^2 = 0$$

$$\Rightarrow r_1 = 0, r_2 = r_3 = -\frac{2}{3}$$

$$\Rightarrow y(x) = C_1 e^{0x} + e^{-\frac{2}{3}x}(C_2 x + C_3)$$
$$= C_1 + e^{-\frac{2}{3}x}(C_2 x + C_3)$$